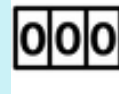
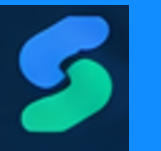


SIDONPAY

Secure Payment. Happier Lives

PAYSIM TRANSACTION ANALYSIS AND FRAUD ANALYSIS DASHBOARD

Insight for Safer Transactions and Smarter Decisions



Total Transactions

6.36M



Total Amount

1.14T



Average Transaction Amount

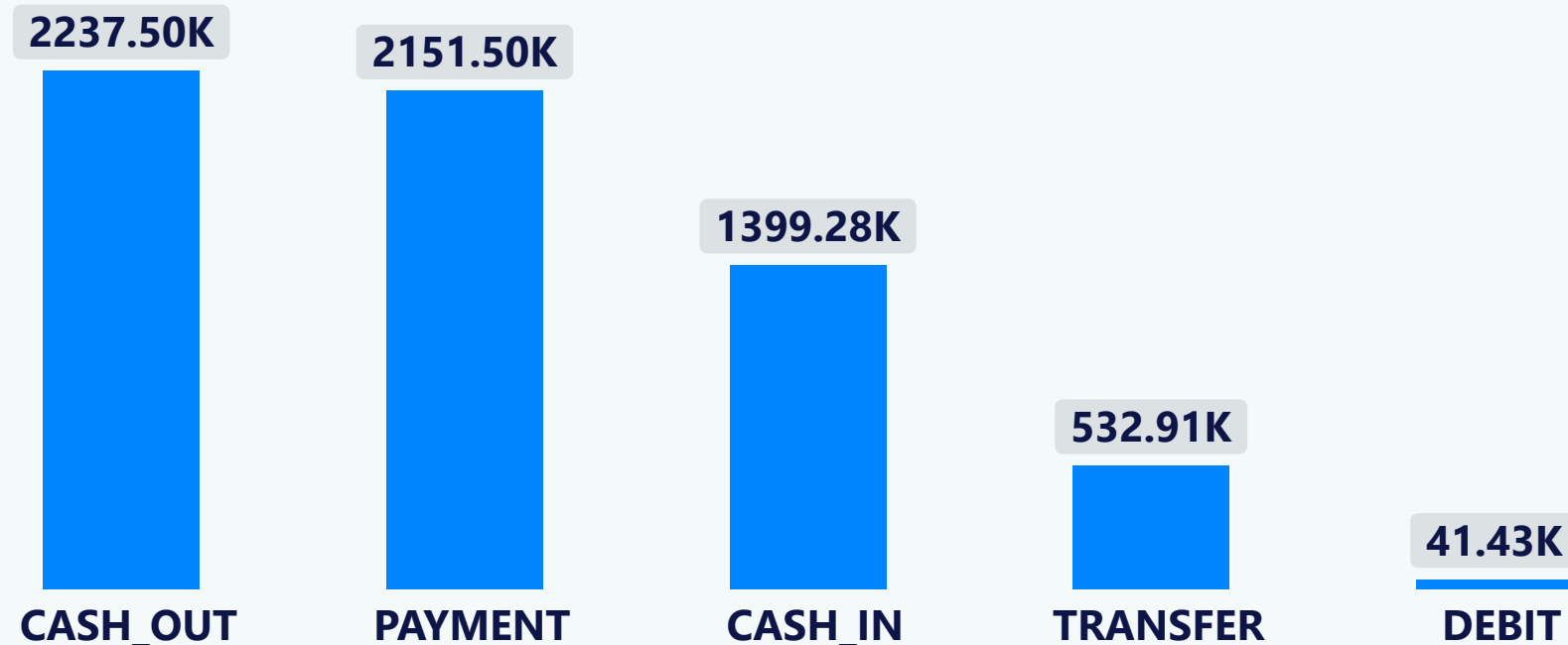
179.86K



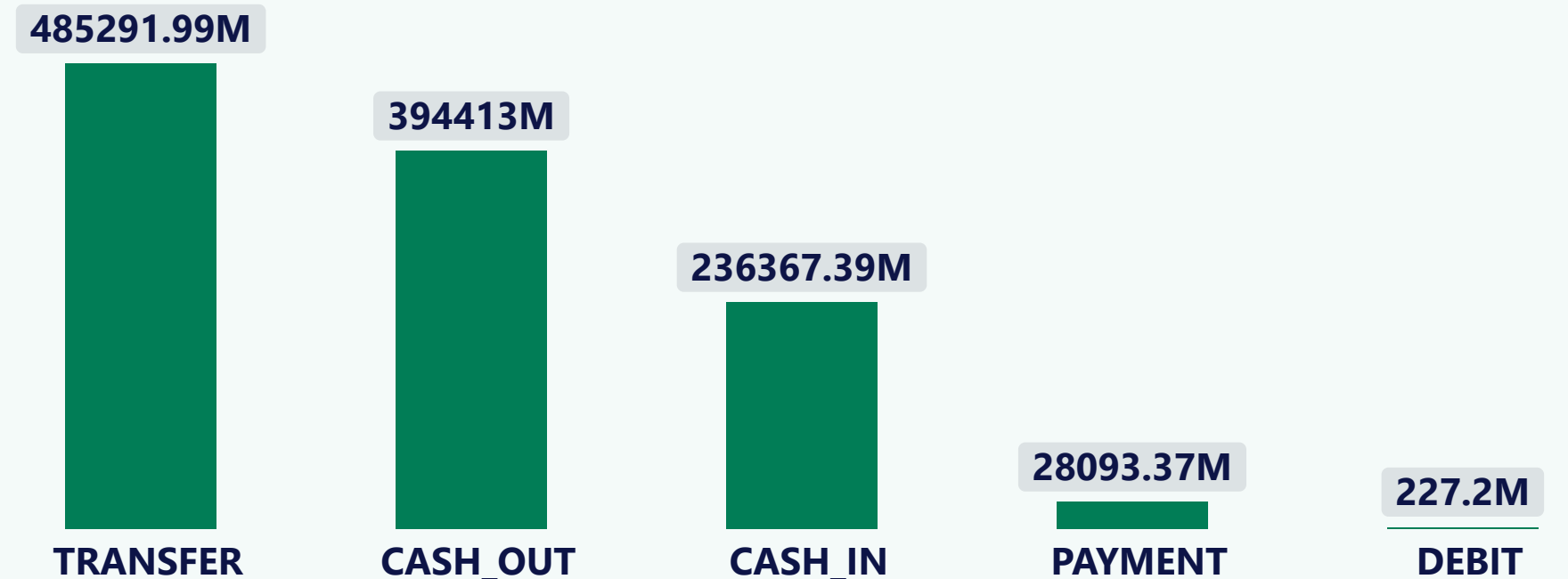
Total Users

6.35M

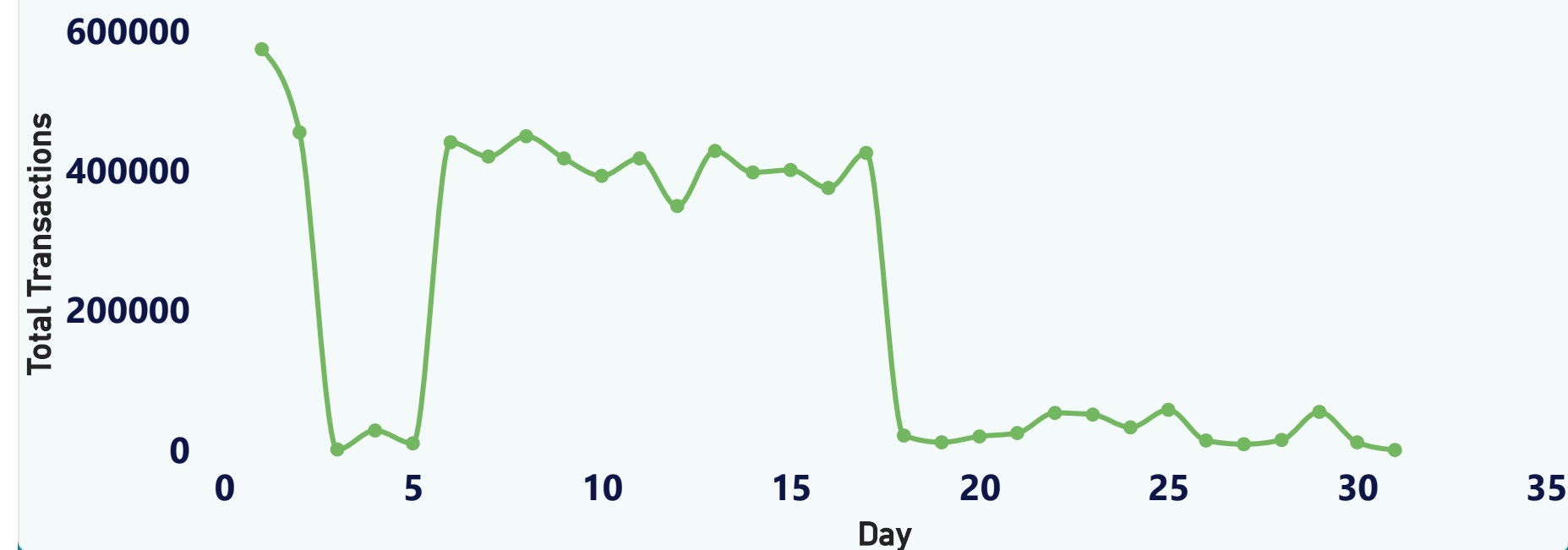
Transactions by type



Transaction Value by type



Transaction Volume Over Time



KEY INSIGHTS

- Cash Out and Payment dominate transaction volume (~4.39M combined).
- Transfer has the highest transaction value despite lower transaction volume.
- Transaction activity peaks around Steps 6–17 before declining sharply.



PAYSIM FRAUD ANALYSIS

Identify where fraud happens and which transaction amount carry the greatest risk



Fraud Transactions

8213



Fraud Rate %

0.13%



Fraud Total Amount

12.06bn

type

All

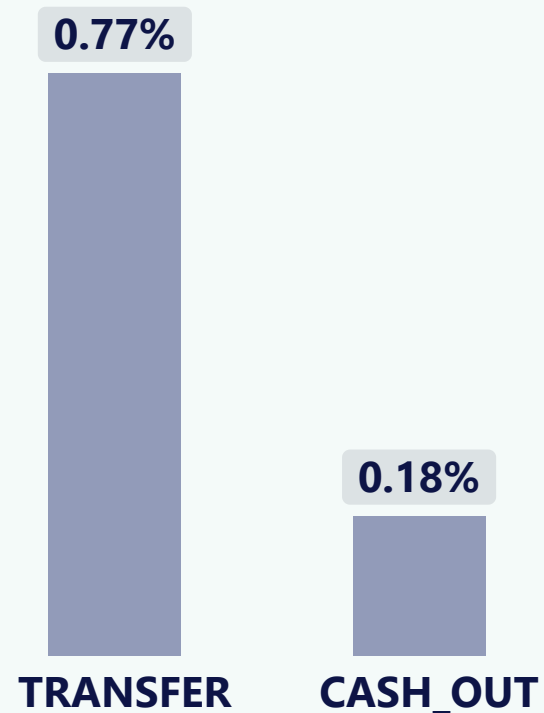
Day

All

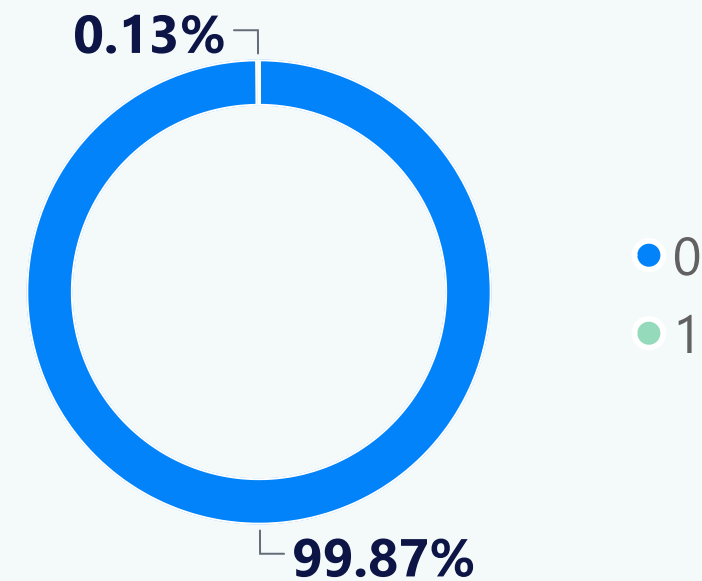
isFraud

All

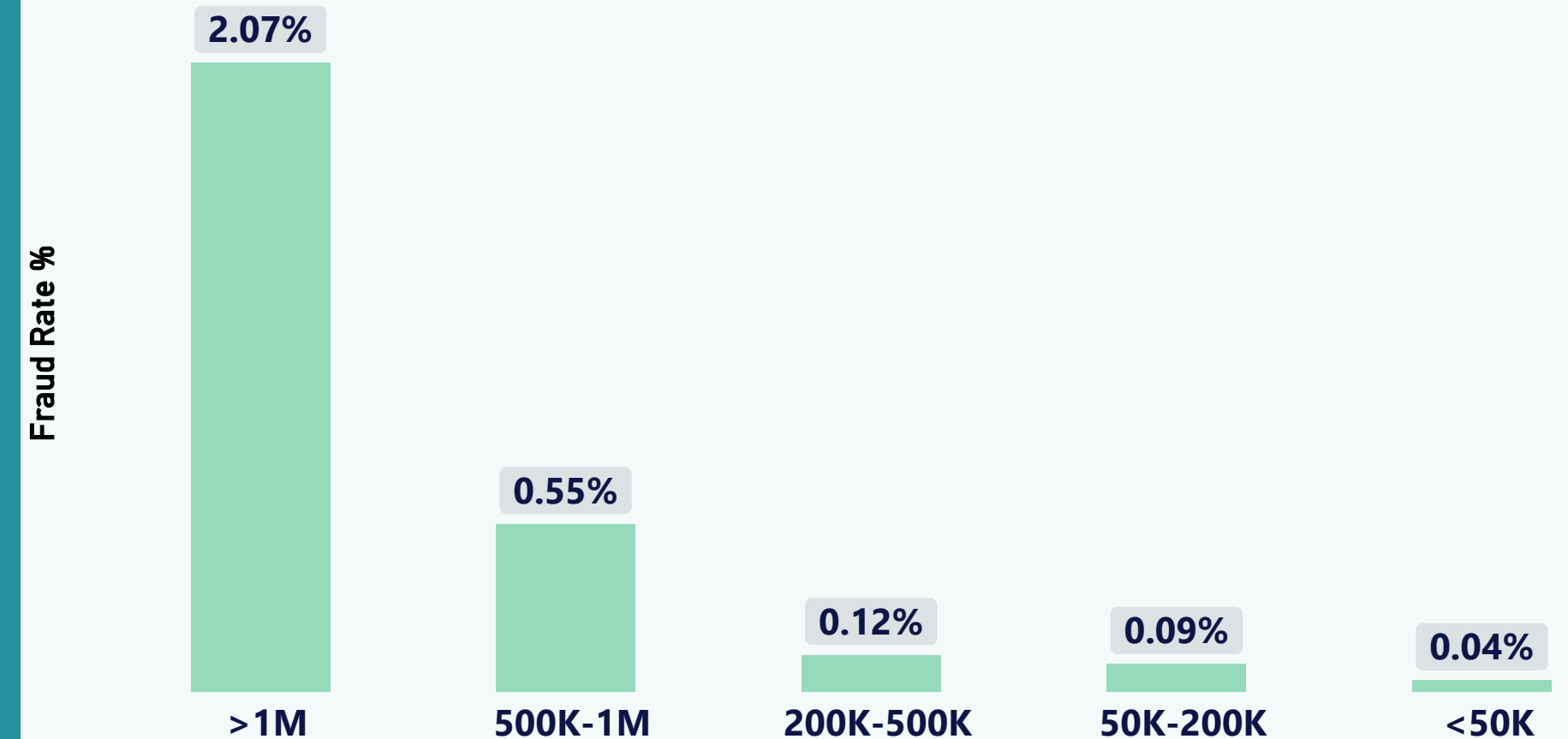
Fraud Rate by type



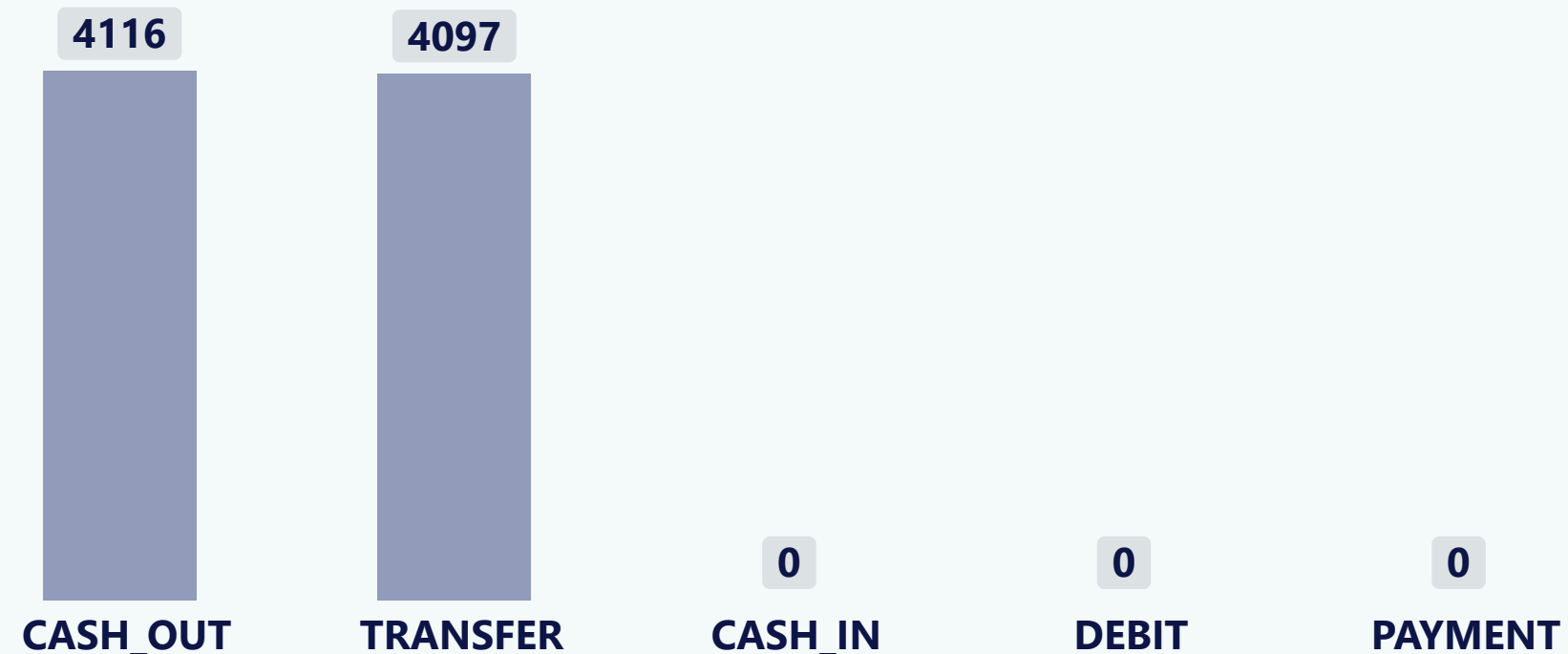
Fraud VS Non Fraud



Fraud Rate by Amount Band



Fraud Transaction by type



💡 KEY INSIGHTS

- Cash Out and Transfer account for all fraud cases (4,116 and 4,097).
- Transfer has the highest fraud rate by type at 0.77%.
- Transactions above 1M have the highest fraud rate (2.07%).
- Overall fraud is low (0.13%), but risk increases sharply with transaction value

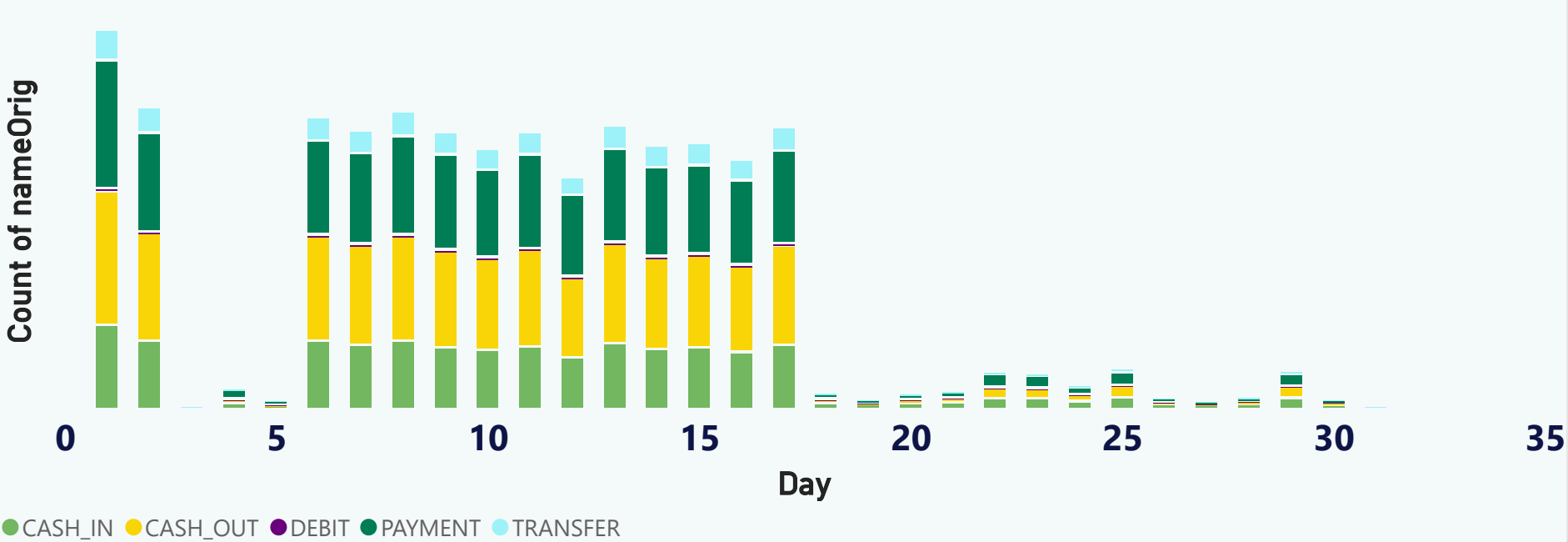


PAYSIM USER AND TRANSACTION BEHAVIOUR

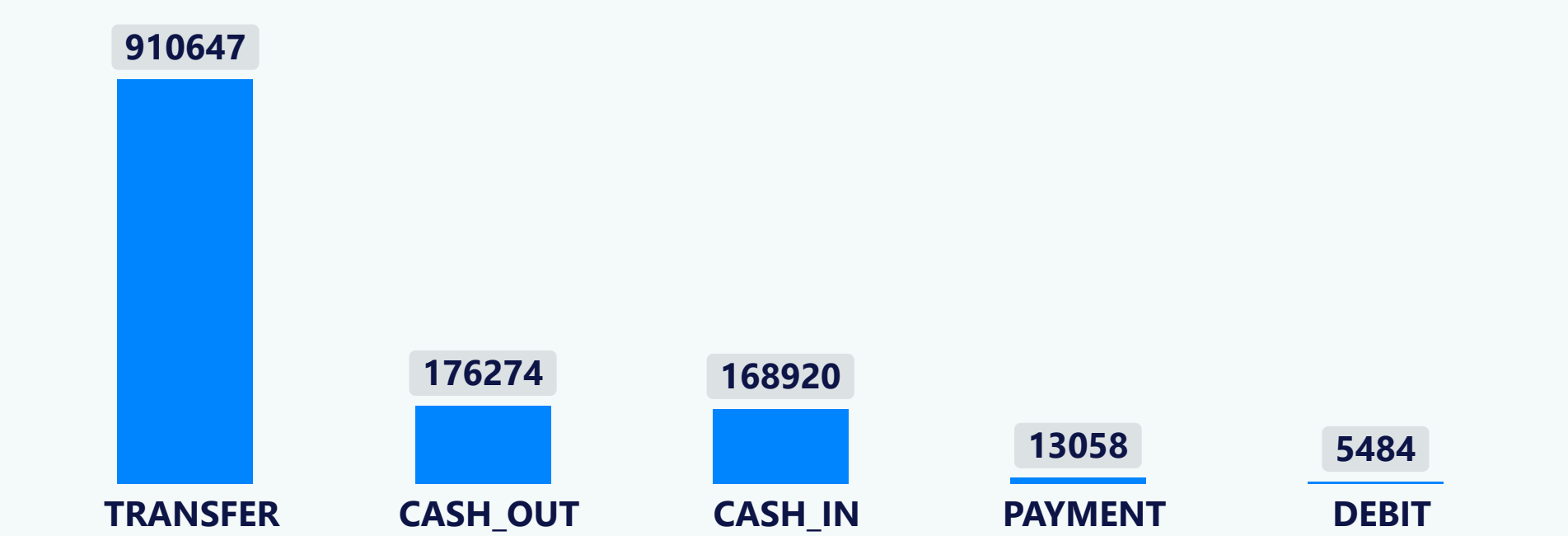
Identify peak and low activity period



Transaction Type Over Time



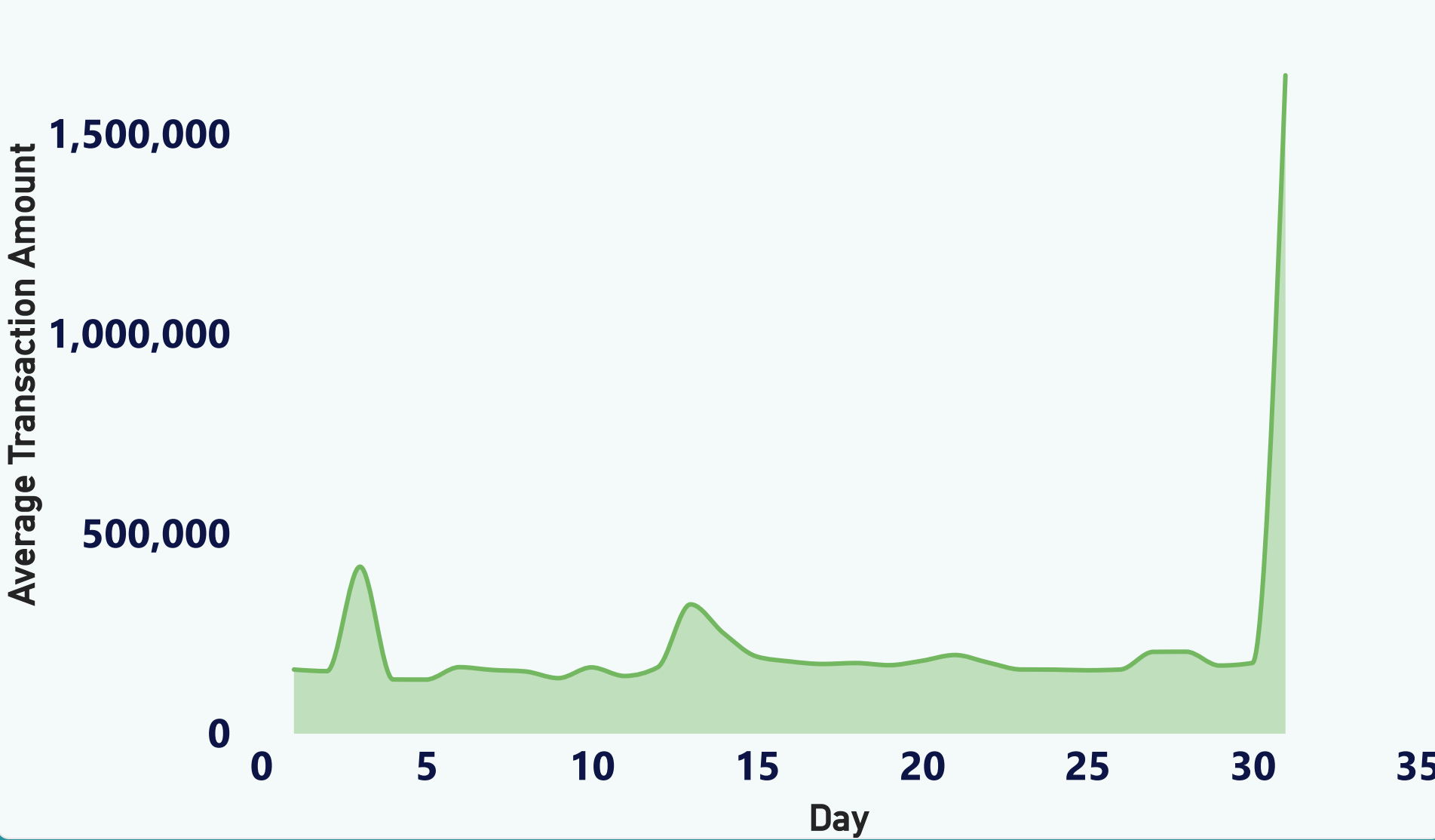
Average Transaction Amount by Type



Top 10 High Frequency User



Average Transaction Amount Over Time



KEY INSIGHTS

- Transfers have the highest average transaction value (910647).
- Transaction activity varies considerably across days.
- A major average-value spike occurs around Day 31, indicating potential high-value activity requiring investigation.



PAYSIM FRAUD DETECTION PERFORMANCE

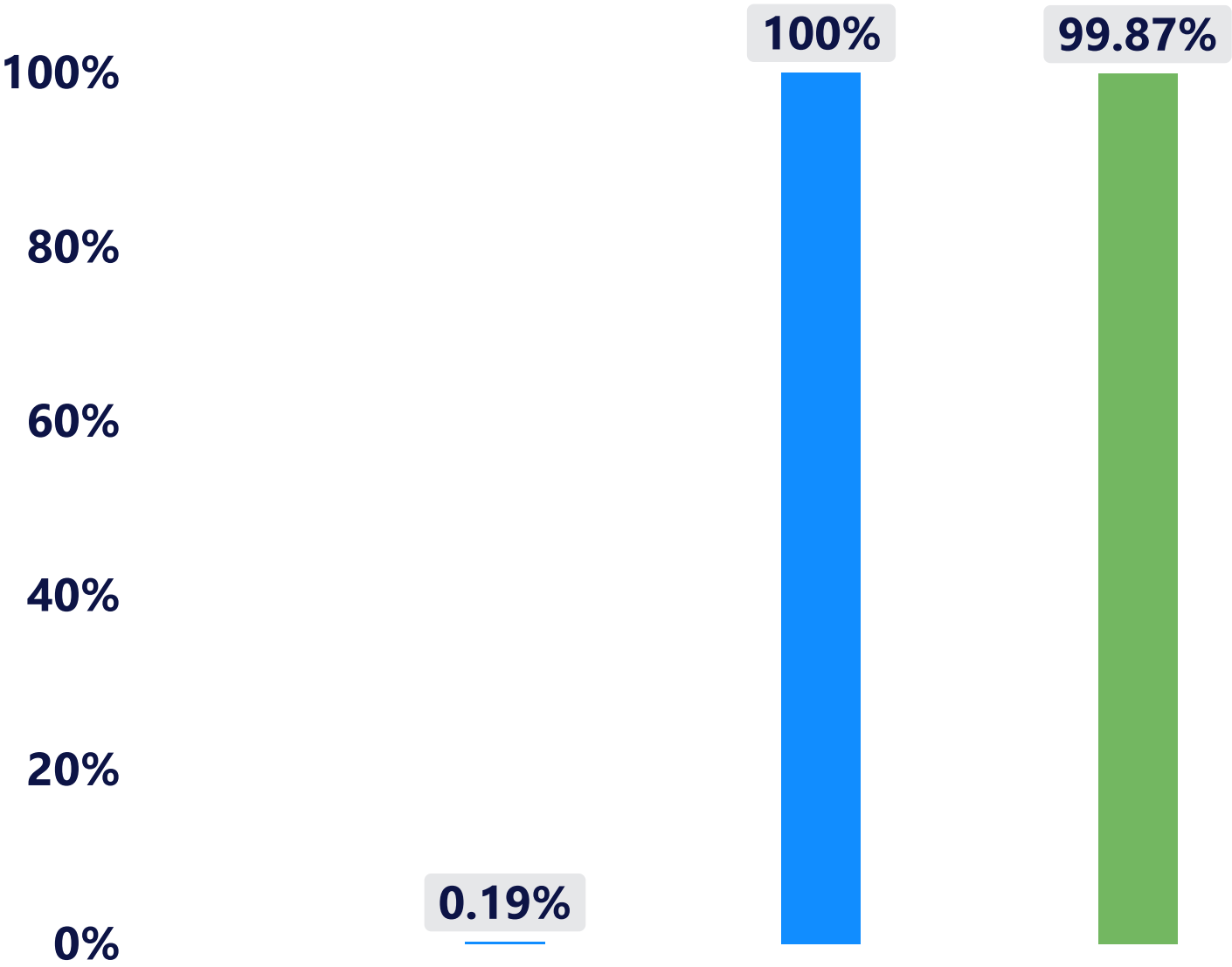
Test whether the current is flagged-fraud mechanism catches the fraud that actually occurs



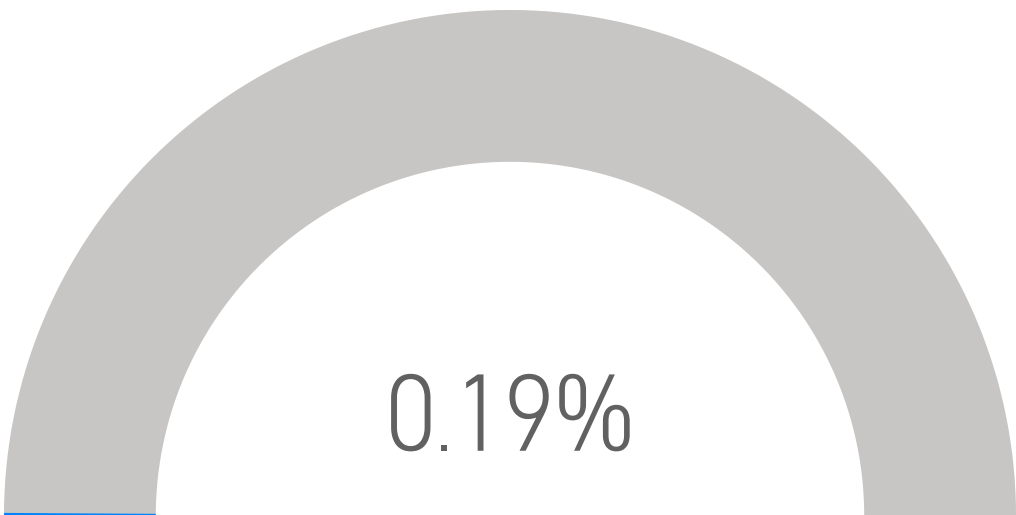
Confusion Matrix

Actual Non-Fraud -	True Negative 6,354,407	False Positive --
Actual Fraud -	False Negative 8197 Fraud Missed by the system	True Positive 16 Fraud Currently Flagged
	Predicted Non-Fraud	Predicted Fraud

Current System Performance

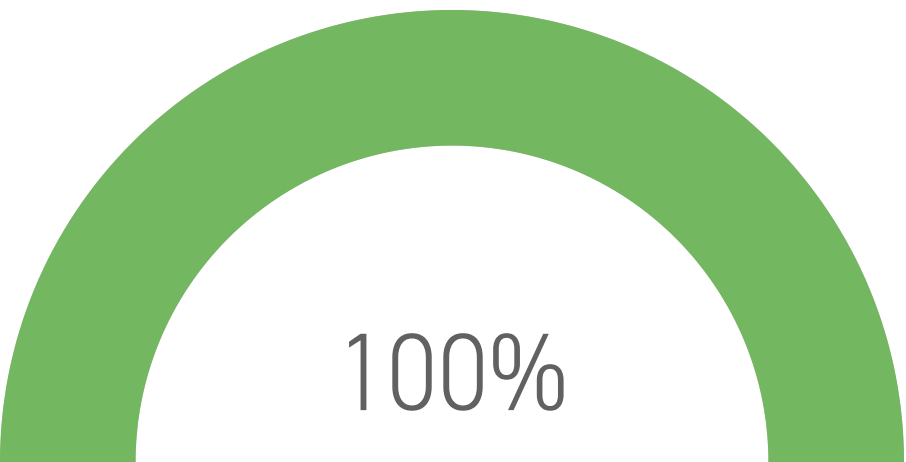


Flag Recall %



Fraud Detection Coverage

Precision



Flagged that were Fraud

● Flag Recall % ● Precision ● Accuracy

KEY INSIGHT.

The current fraud detection system is highly precise but ineffective at detecting fraud. It correctly flags only 16 fraud cases while missing 8197, resulting in just 0.19% recall. The 99.87% accuracy is misleading because fraud represents a very small share of total transactions.



PAYSIM FRAUD MONITORING RECOMMENDATIONS

Translate the analysis into a practical risk monitoring Framework



01 Prioritize Transfer and Cash Out

This two Transaction types account for all 8,213 fraud cases observed in this data set

02 Monitor high - value transaction

The 1M amount band has the highest observed fraud rate: use value as a risk signal, not an automatic fraud decision

03 Move beyond one threshold rule

The current flag captures only 16 of 8213 fraud cases. Combine multiple signals.

04 Build a risk-score workflow

Combine transaction type, amount, unusual behavior, transaction sequence and existing alerts

Proposed Risk Scoring

Transfer	+2
Cash Out	+2
Transfer	+2
> 500	+2
> 1M	+3
Unusual behavior	+3
Transfer -> Cash Out	+3
Existing Flag	+5

0-2 Low | 3-5 Medium
6-8 High | 9+ Critical